Question (6): Gradient of $K = A^T A$ with respect to A

We are given:

1. A is a 2×3 matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}.$$

2. $f(A) = A^T A = K$, where:
   - A^T is the **transpose** of A, and
   - K is the resulting 3×3 matrix.

We need to calculate the **gradient of K with respect to A**, step by step.

Step 1: Understand the Function $f(A) = A^T A$

What is $A^T A$?

- A^T is the transpose of A, where rows and columns are flipped:

$$A^T = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \\ a_{13} & a_{23} \end{bmatrix}.$$

- Multiply A^T (a 3×2 matrix) by A (a 2×3 matrix):

$$K = A^T A = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \\ a_{13} & a_{23} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}.$$

Resulting K:

K is a 3×3 matrix:

$$K = \begin{bmatrix} a_{11}^2 + a_{21}^2 & a_{11}a_{12} + a_{21}a_{22} & a_{11}a_{13} + a_{21}a_{23} \\ a_{12}a_{11} + a_{22}a_{21} & a_{12}^2 + a_{22}^2 & a_{12}a_{13} + a_{22}a_{23} \\ a_{13}a_{11} + a_{23}a_{21} & a_{13}a_{12} + a_{23}a_{22} & a_{13}^2 + a_{23}^2 \end{bmatrix}.$$

Step 2: What Is the Gradient?

The **gradient** of a function maps how the function changes with respect to each variable. For a matrix:

$$\text{Gradient of } K \text{ with respect to } A = \frac{\partial K}{\partial A}.$$

This means we calculate how **each element of K** changes with respect to **each element of A**.

Step 3: Expand the Elements of K

The entries of K are computed as:

$$k_{ij} = \sum_{k=1}^{2} a_{ki}a_{kj}.$$

For example:

- $k_{11} = a_{11}^2 + a_{21}^2$,

- $k_{12} = a_{11}a_{12} + a_{21}a_{22}$,
- $k_{33} = a_{13}^2 + a_{23}^2$.

Step 4: Compute the Gradient

General Formula:

For $K = A^T A$, the derivative of K with respect to A is:

$$\frac{\partial K}{\partial A} = 2A.$$

Let’s understand why.

Proof:

1. Start with $K = A^T A$, where:

$$K_{ij} = \sum_{k=1}^{2} a_{ki}a_{kj}.$$

2. Differentiate k_{ij} with respect to a_{pq}:
   - Case 1: If $p \neq q$, the derivative is 0 because a_{pq} doesn’t appear in k_{ij}.
   - Case 2: If $p = q$, the derivative of k_{ij} is proportional to the corresponding entry in A.
3. Combine the terms to get:

$$\frac{\partial(A^T A)}{\partial A} = 2A.$$

Final Answer

The gradient of $K = A^T A$ with respect to A is:

$$\frac{\partial K}{\partial A} = 2A.$$

This result is valid for any rectangular matrix A.